

1. An airfoil coordinates are provided in the attached Excel file.

1-1: Draw the 2D geometric shape of this airfoil.

1-2: Draw the chord line and the camber line.

1-3: Specify the maximum thickness ratio and the location of the maximum camber.

1-4: Make and draw a wing planform made of this airfoil given:

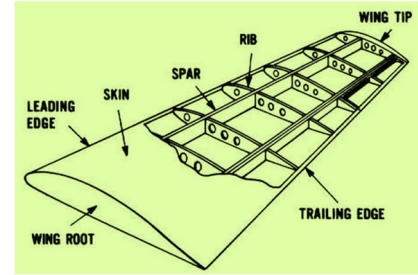
Wing span: 12 m, Root chord: 3 m, Tip chord: 1.2 m, Sweep angle: 35 degrees.

1-5: Draw the mean geometric chord (MGC) of this wing planform.

1-6: Compute the wing mean geometric chord length graphically as well as its coordinates, using both the formulas and the integration scheme introduced in chapter two and compare the results.

1-7: Calculate the zero-lift angle of attack, α_{0_L} for this wing, whose chord and twist distributions are linearly distributed from root to tip as specified below.

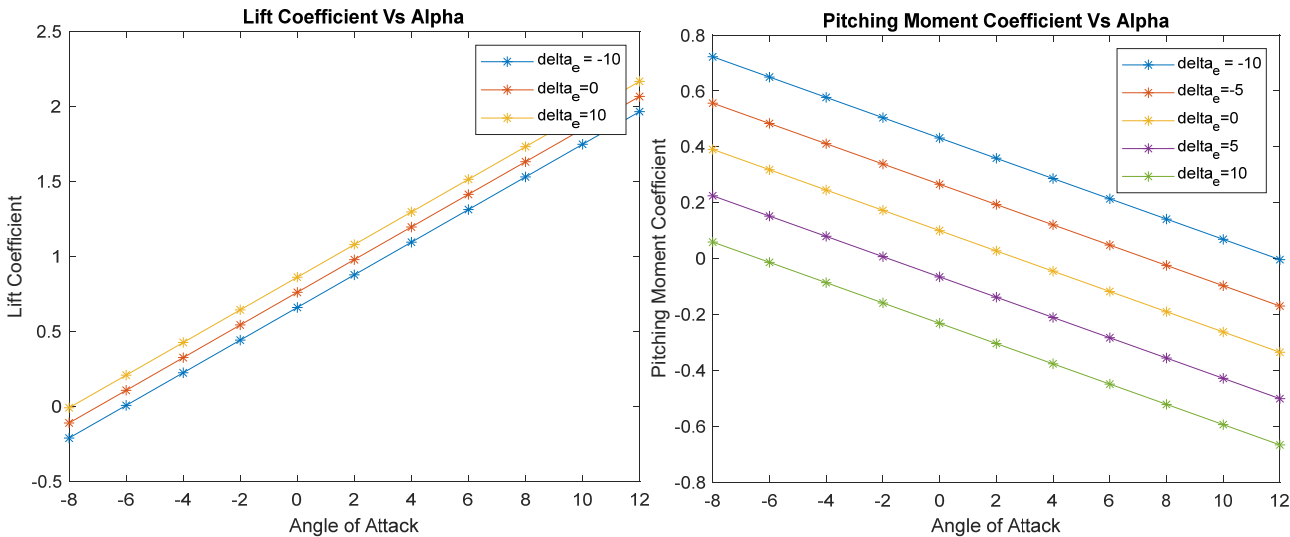
Root chord with $\varepsilon_T = 0, \alpha_{0_L} = -0.5^\circ$; Tip chord with $\varepsilon_T = -1.8^\circ, \alpha_{0_L} = -1.1^\circ$



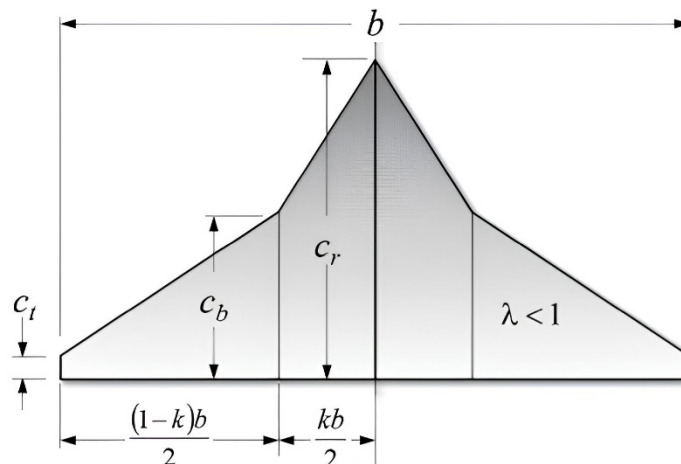
2- Determine the parameters in lift and pitching moment relations given below using the provided graphs.

$$C_L = C_{L_0} + C_{L_\alpha} \alpha + C_{L_{\delta_E}} \delta_E$$

$$C_m = C_{m_0} + C_{m_\alpha} \alpha + C_{m_{\delta_E}} \delta_E$$



3- Consider the double delta wing as shown in the figure:



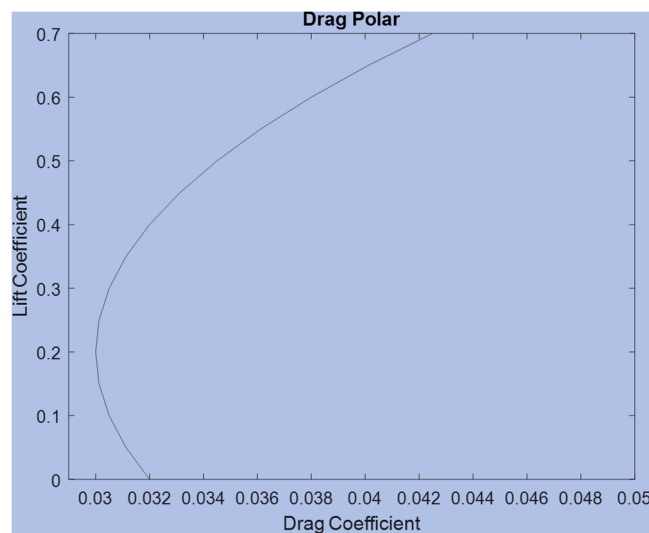
- 3-1 Calculate the wing area.
- 3-2 Calculate aspect ratio for this wing.
- 3-3 Derive an expression for the mean geometric chord.
- 3-4 Simplify the answer for a simple triangular delta wing.
- 3-5 Use F102 wing parameters in your derived formulas to compute its mean geometric chord.
- 3-6 Use F102 3-views, draw the mean geometric chord with the geometric method discussed in the lectures note, and compare its result with what is obtained in 3-5.

4- Refer to appendix B of Roskam's book for airplane A:

- 4-1 Plot the aircraft lift and pitching moment coefficient vs angle of attack and coefficient vs angle of attack for three different elevator settings in climb condition.
- 4-2 Using Prandtl-Glauert transformation (PGT) to correct C_{L_α} for cruise speed, determine the % error due to compressibility effect. Note: Consider the climb C_{L_α} is at zero Mach number to use in PGT.
- 4-3 If the elevator can deflect between -10 to 2 degrees, determine the range of possible angle of attacks and lift coefficients for trimmed flights, assuming the thrust line passes through the aircraft center of gravity.

5- Use the following drag polar curve to for and aircraft,

- 5-1 Extract the zero lift drag coefficient, minimum drag and its corresponding lift coefficient (at minimum drag) respectively,
- 5-2 Determine the k value in the drag polar equation,
- 5-3 If Oswald number is about 0.85, what is the aircraft aspect ratio.



Bonus Problems (Optional, but must be typed if one decides to do it)

B2: An aircraft model is tested in a low-speed wind tunnel at an angle of attack of 20 deg, sideslip of 10 deg, and bank angle of 10 deg. There are load cells (Strain Gauges) within the model that measure the aerodynamic forces acting on the model in the aircraft body coordinate system. The measurements show $F_x = 21.7 \text{ lb}$, $F_y = -33.0 \text{ lb}$, and $F_z = -91 \text{ lb}$. Determine:

- 1- The body to wind tunnel transformation matrix, T_b^w .
- 2- The lift, drag, and side forces acting on the model aircraft.